

Quarto Grid Challenge

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Introduction

Long bones develop through the process of Endochondral Ossification (EO) [1], [2], [3]. This process starts with the condensation of mesenchymal cells, continuing with intricate processes of differentiation, proliferation, hypertrophy, apoptosis and matrix modification [1], [2], [3]. During this transformation, hyaline cartilage is replaced by bone, developing primary (POC) and secondary (SOC) ossification centers; the former allows longitudinal development, while the latter promotes epiphysis formation. This progression is modulated by multiple factors, including cellular, biochemical, and mechanical effects [4], [5]. Disturbances on this growth mechanism can cause different pathologies in children and teenagers; therefore, understanding this process is crucial to explain, prevent and treat these disorders [6], [7]. For this reason, several mathematical and computational models have been proposed to simulate EO and SOC emergence. Early works suggested that cyclical mechanical stimuli could influence ossification, allowing the prediction on location of SOCs [4].

Additionally, other factors have been considered, e.g., bone remodelling [8], [9], biochemical feedback loops and cellular mechanisms [10], [11], [12], or a combination of these elements [13], [14], [15], [16]. Given that multiple SOCs can appear on a single epiphysis [7], studies based on reaction-diffusion modeling have been conducted to analyze the influence of epiphyseal size on the number of SOCs [17]. However, previous models have mostly focused on the emergence of a single SOC, overlooking multiple SOCs presence on the epiphysis, as observed in the humerus, tibia, femur, and other bones. Therefore, this study builds on previous work that predicted SOC locations based primarily on geometric and mechanical factors [18] by incorporating biochemical influences. It aims to explore how geometry, mechanics, and biochemical conditions interact to govern SOC formation, including double-center ossification cases.

- [1] H. M. Kronenberg, "Developmental regulation of the growth plate." *Nature*, vol. 423, no. 6937, pp. 332–336, May 2003, doi: [10.1038/nature01657](https://doi.org/10.1038/nature01657).
- [2] P. Aghajanian and S. Mohan, "The art of building bone: Emerging role of chondrocyte-to-osteoblast transdifferentiation in endochondral ossification," *Bone Research*, vol. 6, no. 1, p. 19, Jun. 2018, doi: [10.1038/s41413-018-0021-z](https://doi.org/10.1038/s41413-018-0021-z).
- [3] J. C. Nguyen, B. K. Markhardt, A. C. Merrow, and J. R. Dwek, "Imaging of Pediatric Growth Plate Disturbances," *RadioGraphics*, vol. 37, no. 6, pp. 1791–1812, Oct. 2017, doi: [10.1148/rg.2017170029](https://doi.org/10.1148/rg.2017170029).
- [4] D. R. Carter and M. Wong, "The role of mechanical loading histories in the development of diarthrodial joints," *Journal of Orthopaedic Research*, vol. 6, no. 6, pp. 804–816, 1988, doi: [10.1002/jor.1100060604](https://doi.org/10.1002/jor.1100060604).
- [5] S. Sundaramurthy and J. J. Mao, "Modulation of endochondral development of the distal femoral condyle by mechanical loading." *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*, vol. 24, no. 2, pp. 229–241, Feb. 2006, doi: [10.1002/jor.20024](https://doi.org/10.1002/jor.20024).
- [6] E. Comellas and S. J. Shefelbine, "The role of computational models in mechanobiology of growing bone," *Frontiers in Bioengineering and Biotechnology*, vol. 10, 2022, doi: [10.3389/fbioe.2022.973788](https://doi.org/10.3389/fbioe.2022.973788).
- [7] J. A. Ogden, R. N. Hensinger, and N. McCollough, "Diagnostic Imaging," in *Skeletal injury in the child*, 3rd ed., Springer, 2000, p. 128.
- [8] M. Wong and D. R. Carter, "A theoretical model of endochondral ossification and bone architectural construction in long bone ontogeny." *Anatomy and embryology*, vol. 181, no. 6, pp. 523–532, 1990, doi: [10.1007/BF00174625](https://doi.org/10.1007/BF00174625).

- 29 [9] S. S. Stevens, G. S. Beaupré, and D. R. Carter, “Computer model of endochondral growth and ossification in long bones: Biological and mechanobiological influences.” *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*, vol. 17, no. 5, pp. 646–653, Sep. 1999, doi: [10.1002/jor.1100170505](https://doi.org/10.1002/jor.1100170505).
- 30 [10] D. A. Garzón-Alvarado, J. M. García-Aznar, and M. Doblaré, “A reaction-diffusion model for long bones growth.” *Biomechanics and modeling in mechanobiology*, vol. 8, no. 5, pp. 381–395, Oct. 2009, doi: [10.1007/s10237-008-0144-z](https://doi.org/10.1007/s10237-008-0144-z).
- 31 [11] D. A. Garzón-Alvarado, J. Garca-Aznar, and M. Doblaré, “Appearance and location of secondary ossification centres may be explained by a reaction–diffusion mechanism,” *Computers in biology and medicine*, vol. 39, no. 6, pp. 554–561, 2009.
- 32 [12] J. Stoop, Y. Yokoyama, and T. Adachi, “Timing of resting zone parathyroid hormone-related protein expression affects maintenance of the growth plate during secondary ossification: A computational study,” *Biomechanics and Modeling in Mechanobiology*, vol. 24, no. 1, pp. 125–137, Feb. 2025, doi: [10.1007/s10237-024-01899-3](https://doi.org/10.1007/s10237-024-01899-3).
- 33 [13] L. M. Peinado Cortés, J. C. Vanegas Acosta, and D. A. Garzón Alvarado, “A mechanobiological model of epiphysis structures formation,” *Journal of Theoretical Biology*, vol. 287, pp. 13–25, 2011, doi: [10.1016/j.jtbi.2011.07.011](https://doi.org/10.1016/j.jtbi.2011.07.011).
- 34 [14] J. J. Vaca-González, M. Moncayo-Donoso, J. Guevara, Y. Hata, S. J. Shefelbine, and D. A. Garzón-Alvarado, “Mechanobiological modeling of endochondral ossification: An experimental and computational analysis,” *Biomechanics and modeling in mechanobiology*, vol. 17, no. 3, pp. 853–875, Jan. 2018, doi: [10.1007/s10237-017-0997-0](https://doi.org/10.1007/s10237-017-0997-0).
- 35 [15] A. F. Carrera-Pinzón, K. Márquez-Flórez, R. H. Kraft, S. Ramtani, and D. A. Garzón-Alvarado, “Computational model of a synovial joint morphogenesis,” *Biomechanics and Modeling in Mechanobiology*, vol. 19, no. 5, pp. 1389–1402, Oct. 2020, doi: [10.1007/s10237-019-01277-4](https://doi.org/10.1007/s10237-019-01277-4).
- 36 [16] S. M. Sadeghian, F. D. Shapiro, and S. J. Shefelbine, “Computational model of endochondral ossification: Simulating growth of a long bone,” *Bone*, vol. 153, p. 116132, 2021, doi: [10.1016/j.bone.2021.116132](https://doi.org/10.1016/j.bone.2021.116132).
- 37 [17] D. A. Garzón-Alvarado, “Can the size of the epiphysis determine the number of secondary ossification centers? A mathematical approach,” *Computer Methods in Biomechanics and Biomedical Engineering*, vol. 14, no. 9, pp. 819–826, 2011.
- 38 [18] C. R. Bustamante-Porras, K. Marquez-Florez, C. A. Duque-Daza, and D. A. Garzón-Alvarado, “Modeling endochondral ossification: Effects of mechanical loading and bone shape,” *Journal of Orthopaedics*, vol. 68, pp. 197–218, Oct. 2025, doi: [10.1016/j.jor.2025.05.034](https://doi.org/10.1016/j.jor.2025.05.034).